Investment in New Technology

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Why This Project

- AI technologies behave like a new type of capital.
- Firms face a dynamic problem:
  invest AI today to improve productivity tomorrow.
- A key economic parameter—the elasticity of substitution—controls how easily AI can replace labor.
- Incentives to invest in AI capital may vary with different elasticity.

Our Goal: Explores how investment decisions respond to different elasticity values.

Research Question

- How does the elasticity of substitution affect a firm's optimal investment in AI capital?
- How do AI capital, output, consumption, and investment paths differ under elasticity = 1.05, 1.2, and 1.3?
- How does the long-run steady state of AI capital respond to changes in elasticity?

Model: State and Action

State (xt): What the firm knows at time t

$$x_t = A_t$$

- At: AI capital stock at time t

Action (at): The firm's decision

$$a_t = I_t$$

- Action is choosing investment in AI capital.
- Consumption is whatever left:

$$C_t = Y_t - I_t$$

Model: Production

CES Production Function:

$$Y_t = \left[\frac{1}{2}L^{\frac{\sigma-1}{\sigma}} + \frac{1}{2}A_t^{\frac{\sigma-1}{\sigma}}\right]^{\frac{\sigma}{\sigma-1}}$$

- Labor is fixed at L = 1.
- σ: elasticity of substitution between labor and AI
- If σ is greater than 1, AI is a substitute for labor

Model: Policy Function

- **Policy Function:**
  The policy gives the optimal **next-period AI capital** for each current capital level:

$$At + 1 = \pi(At)$$

- **Meaning:**
  The agent chooses $At + 1$ to maximize

$$u(Ct) + \beta V(At + 1)$$

  This is exactly the choice variable used in the code and in the Bellman equation.

- **Consumption implied by the policy:**

$$Ct = Yt + (1 - \delta A)At - At + 1$$

- **Interpretation:**
  Higher At often leads to a higher optimal $At + 1$.
  The policy function summarizes the best dynamic choice in every state.

Model: Reward and Utility

- **Reward Function (per-period utility):**

$$R(xt, at) = u(Ct)$$

- Where $at = It$ is the action
- Reward is utility from consumption in period t
- **Utility Function:**

$$u(Ct) = log(Ct)$$

- Diminishing marginal utility of consumption
- Encourages smoothing between today's consumption and future capital

Model: State Transition

- **Law of Motion for AI Capital:**

$$At + 1 = at$$

This is because the action *is* the choice of next-period capital.

- **Investment:**

$$It = At + 1 - (1 - \delta A)At$$

- **Interpretation:**
  - Choosing a larger $At + 1$ means higher investment
  - If $At + 1 = (1 - \delta A)At$, investment is zero
  - The model is dynamic because the action directly sets tomorrow's state

Full Bellman Equation

- **Value Function:**

$$V(A_t) = \max_{I_t \geq 0} \log(Y_t - I_t) + \beta V(A_{t+1})$$

- **Subject to:**

$$Y_t = \left[\frac{1}{2} L^{\frac{\sigma-1}{\sigma}} + \frac{1}{2} A_t^{\frac{\sigma-1}{\sigma}} \right]^{\frac{\sigma}{\sigma-1}}$$

$$A_{t+1} = A_t(1 - \delta_A) + I_t.$$

- **Intertemporal trade off**:
  Consuming output today OR investing in AI to raise future production.

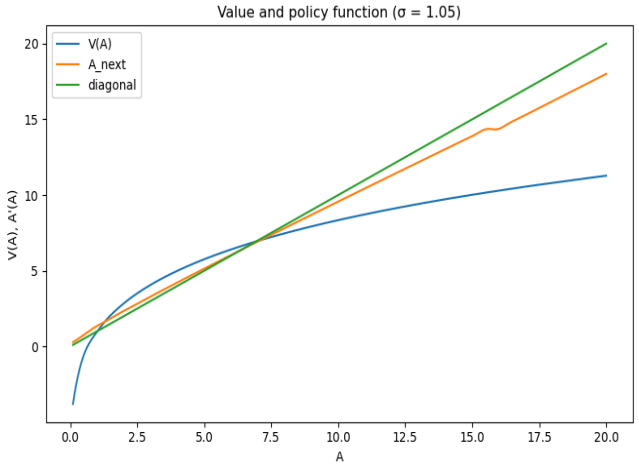
Simulation Setup

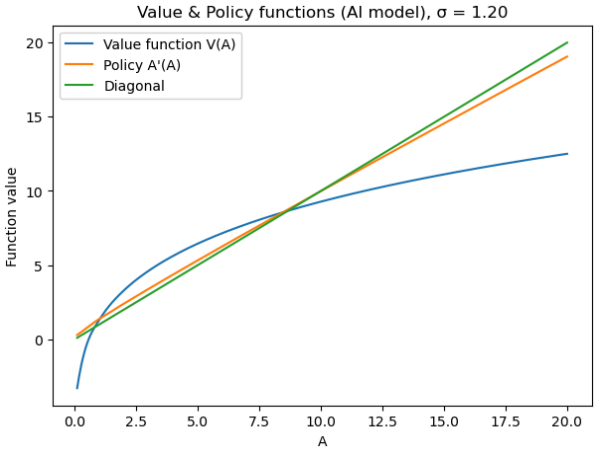
- **Parameters** $\beta = 0.9, \qquad \delta_A = 0.1$

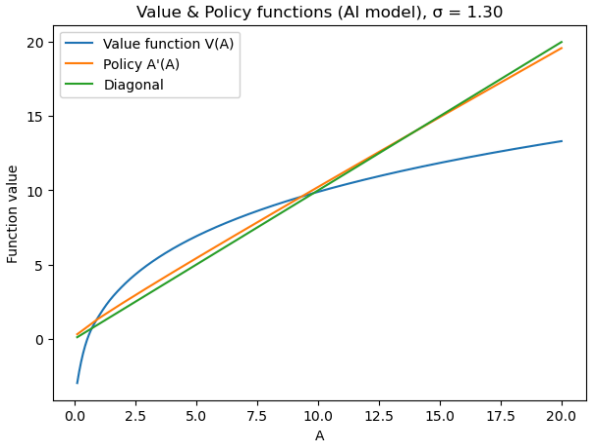
- **Simulate for** $\sigma \in \{1.05,\ 1.2,\ 1.3\}$

- **Initial AI Capital Conditions** $A_0 \in \{0.1,\ 20,\ 10.2531\}$

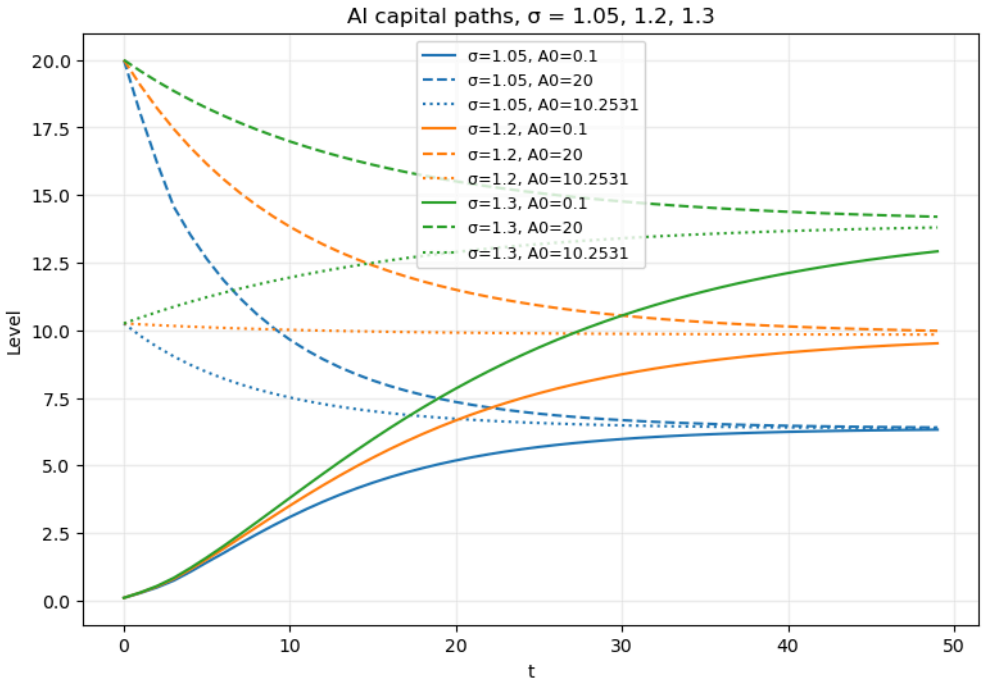
Policy & Value Function





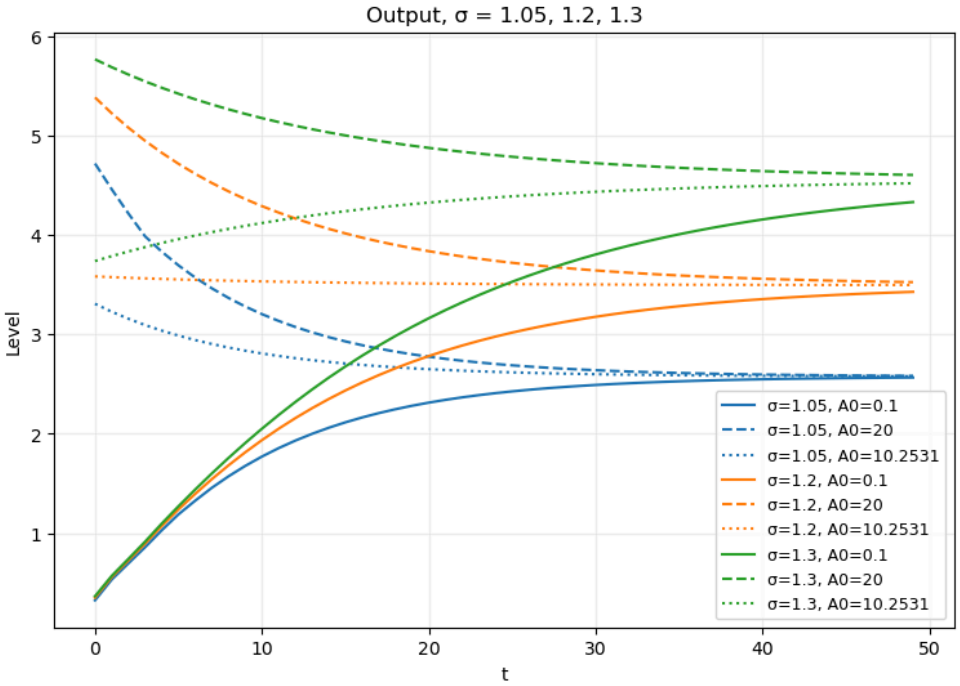


Simulated AI Capital Paths



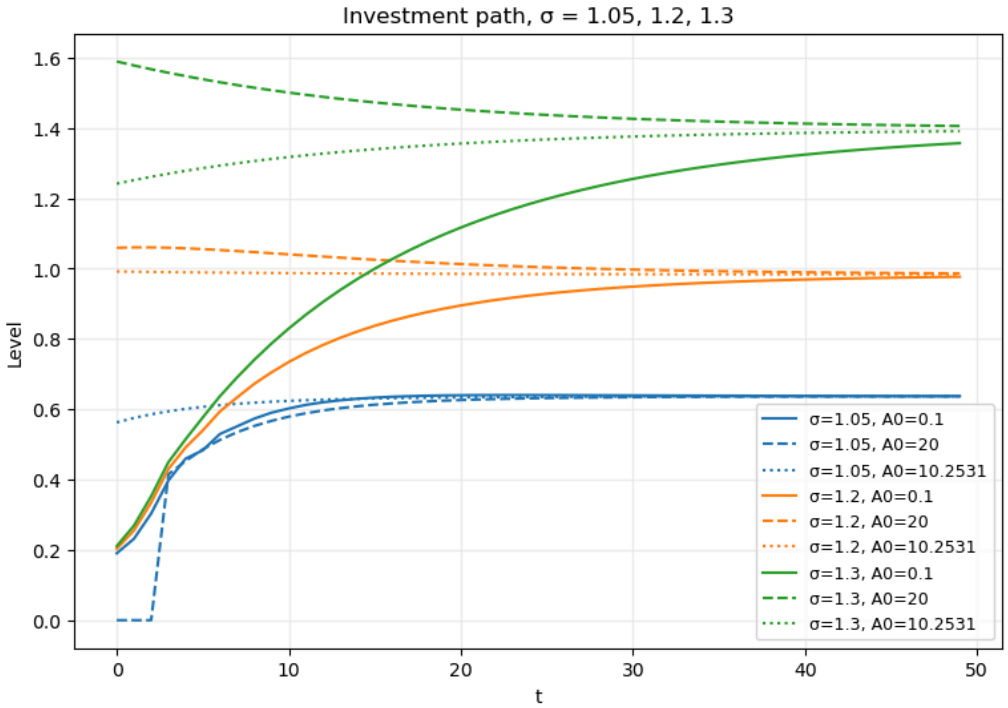
- Low elasticity (1.05): capital approaches a lower long-run level.
- Medium elasticity (1.2): capital approaches an intermediate level.
- High elasticity (1.3): capital approaches a higher level.
- Higher elasticity strengthens AI capital accumulation in our simulations.

Simulated Output Paths



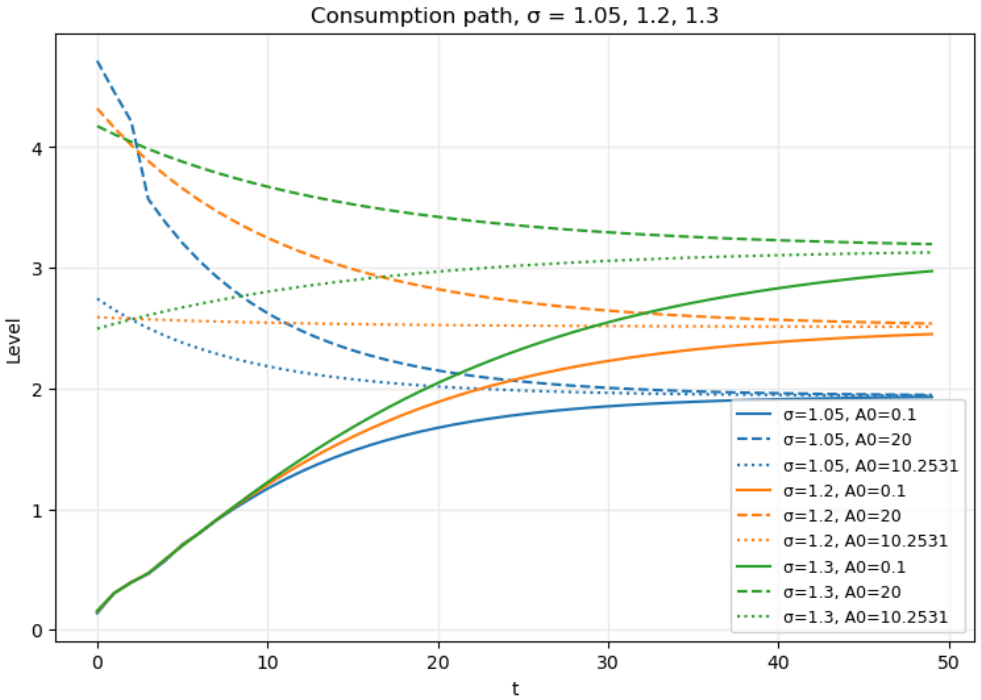
- Higher elasticity produces higher output in our simulations.
- Initial AI capital shapes the transition path.
- Easier AI-labor substitution raises output with fixed labor.
- From low initial capital, output rises toward its long-run level.

Simulated Investment Paths



- Higher elasticity supports higher sustained investment in our simulations.
- Initial AI capital shapes the investment path.
- Easier AI-labor substitution strengthens the incentive to invest in AI capital.

Simulated Consumption Paths



- Higher elasticity produces higher consumption in our simulations.
- Initial AI capital shapes the consumption path.
- Easier AI-labor substitution supports higher consumption as capital accumulates.

Main Takeaways

- Capital paths from different starting values move toward a common level within each elasticity case.
- Easier AI-labor substitution strengthens investment incentives.
- Higher elasticity supports greater capital accumulation, output and consumption in our simulations.
- The model highlights the consumption benefits of more substitutable AI capital.

Limitations

- Labor is fixed and uniform. The model cannot estimate employment or wage changes across different jobs.
- Parameters are illustrative. Real-world AI efficiency and adoption speed may differ from the model.
- Three starting values and 50 periods do not prove global stability or exact steady states. Investment need not be higher at every date.
- Results depend on the grid and interpolation. A full welfare ranking requires comparing discounted values, not just consumption endpoints.

Thank you!
Q & A?